As against their biological counterparts, computer viruses do not evolve on their own. They have to be specifically created by a programmer or virus-creating software. Contrary to popular perception, bugs in a program cannot morph into viruses, though they can leave problems that virus makers (or viruses) can exploit.

Viruses can be divided into two types; on the basis of their behaviour when they are executed: Resident and Non-resident viruses.

Non-resident viruses immediately search for other hosts that can be infected; infect these targets, and finally transfer control to the application program they infected.

Resident viruses do not search for hosts when they are run. Instead, a resident virus loads itself onto memory upon execution and transfers control to the host program. The virus stays active in the background and infects new hosts when those files are accessed by other programs or the operating system itself.

In order to avoid detection, viruses employ different ways of hiding or camouflaging themselves. Older viruses, especially those working on the MS-DOS platform, made sure that the "last modified" date of a host file remained the same when the file is infected by the virus. Some viruses can infect files without increasing their sizes or damaging the files. They accomplish this by overwriting unused areas of executable files. These are called cavity viruses.

Recent viruses avoid any kind of detection by forcefully killing the tasks associated with the virus scanner before it can detect them. Other viruses try to fool anti-virus software by intercepting its requests to the OS. This method is known as the stealth method. A virus can hide itself by ensuring that a request of anti-virus software to read an infected file is passed to the virus, instead of to the OS. The virus can then return an uninfected version of the file to the anti-virus software, so that it seems that the